

risks/issues, conduct detection and monitoring analysis, and implement new processes. “They need to be able to construct analysis mechanisms, provide analysis data, and help integrate related requirements and material selection,” says IDC’s Chiang. “Overall, they spend most of their time focussing on the variation of quality, and control/manage the balance between quality, output and costs.”

The foundry ecosystem offers a multidisciplinary set of jobs to everyone, from diploma holders to engineering graduates and students with science-related qualifications. “Responsibilities of a manufacturing technician can be handled by diploma holders who have proper training of about six months in fab operations. The process, quality assurance and equipment engineers require an engineering degree with a specialisation,” says Satya Gupta, President of VLSI Society of India.

Sachin Alug, CEO of digital talent solutions firm NLB Services, says that VLSI companies may offer anything between ₹10-20 lakh per annum to entry-level professionals, while aggregate salaries offered by MNCs to freshers is between ₹30-80 lakh per annum.

courses in the related fields. For instance, IIT Kanpur offers a course on IC fabrication, and various courses related to electronics device physics & modelling, and circuits. “Additionally, the Material Science and Engineering (MSE) and Chemical Engineering departments offer courses related to semiconductor fabrication. The MSE department has a course on packaging as well,” says S. Sundar Kumar Iyer, Professor in the Department of Electrical Engineering at IIT Kanpur. Separately, IIT Madras has launched a four-year online Bachelor of Science in Electronic Systems programme.

As India focusses on becoming a chip-making nation, the All India Council for Technical Education (AICTE) has also designed a curriculum for BTech in Electronics (VLSI Design and Technology) and Diploma in IC manufacturing. The BTech course covers a range of topics, from IC design and technology and semiconductor device fundamentals to VLSI and analog IC design and physics of electrical engineering materials, among others.

Alternatively, the diploma course covers an introduction to VLSI Fabrication, semiconductor-fab familiarisation, electronics devices and circuits, clean room technologies, semiconductor technology equipment maintenance, along with allied activities in a foundry, such as safety protocols, vacuum technology, etc. With-in industrial automation, the course covers chip packaging and testing, electronics system assembly or product design and renewable energy technologies.

Despite the availability of these courses in the country, a common refrain among industry leaders is that the faculty who would teach these subjects, themselves

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With over 2 million engineering students graduating every year in the country, and only close to 300,000 high-paying jobs available, semiconductors is one buzzing industry, where even the government and universities are seeing significant potential. As the government drives India Inc. to step up investments in the sector, several premier engineering institutes have introduced

A FOUNTAIN OF KNOWLEDGE

Students enrolled in engineering courses at undergraduate level across India in 2021, by discipline

Discipline	Computer Science	Mechanical	Electronics	Civil	Electrical
Students enrolled	1,081.68	595.41	587.81	447.79	354.69

IN THOUSANDS; SOURCE: STATISTA

MARKET METRICS

Globally, around \$500 billion is expected to be invested between 2021 and 2023 in setting up 84 chip-making facilities

SEMICONDUCTOR MARKET

- 2021 → \$600 billion
- 2030 → \$1 trillion*

NEW FAB CONSTRUCTIONS AROUND THE WORLD

- 2021 → 23
- 2022 → 33
- 2023 → 28*

*ESTIMATES; SOURCE: MCKINSEY, WORLD FAB FORECAST REPORT BY SEMI